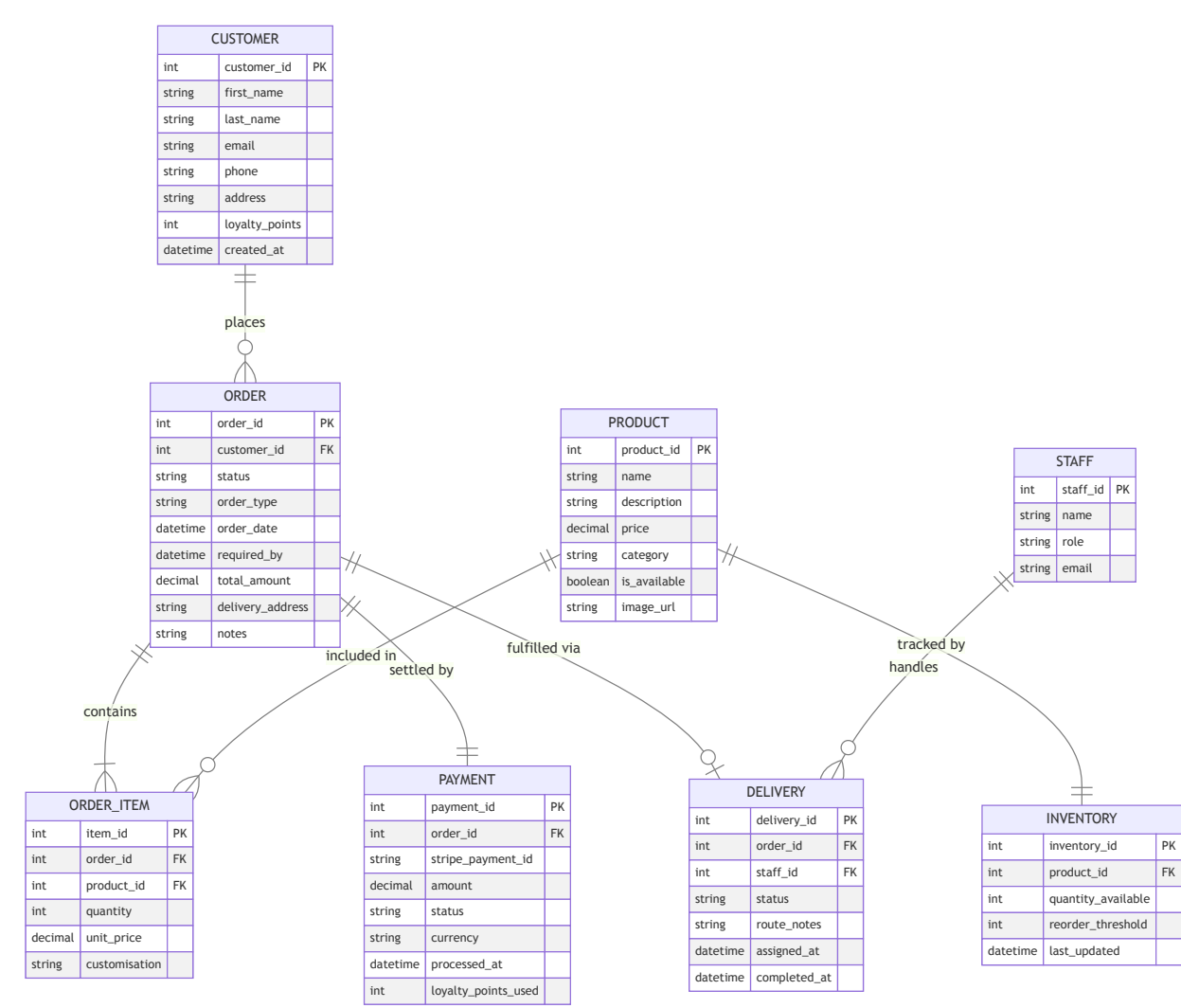


Entity Relationship Diagram (ERD)

FreshBakes Bakery | IS501 Project

Core Entities: Customer · Order · Product · Inventory · Payment **Supporting Entities:** Order_Item · Delivery · Staff



Entity Descriptions

Entity	Description	Key Attributes
CUSTOMER	Registered user who places orders and earns loyalty points	customer_id, email, loyalty_points
ORDER	A customer's placed order, tracking type and fulfilment status	order_id, status (Pending→Delivered), order_type

Entity	Description	Key Attributes
		(delivery/collection)
ORDER_ITEM	Line item linking an order to a specific product with quantity and any customisation notes	item_id, quantity, customisation
PRODUCT	A bakery item available for purchase, with availability flag	product_id, price, is_available
INVENTORY	Tracks current stock and reorder threshold for each product	quantity_available, reorder_threshold
PAYMENT	Records the Stripe transaction tied to an order, including loyalty points applied	stripe_payment_id, amount, loyalty_points_used
DELIVERY	Assignment of a delivery-type order to a staff member, with route notes and completion timestamp	delivery_id, status, route_notes
STAFF	Baker or delivery employee with an assigned role	staff_id, role

Relationships

Relationship	Cardinality	Meaning
CUSTOMER → ORDER	One-to-many	A customer may place many orders
ORDER → ORDER_ITEM	One-to-many (mandatory)	Every order contains at least one line item
PRODUCT → ORDER_ITEM	One-to-many	A product can appear in many order items
PRODUCT → INVENTORY	One-to-one	Each product has exactly one inventory record
ORDER → PAYMENT	One-to-one	Every confirmed order has exactly one payment record
ORDER → DELIVERY	One-to-zero-or-one	Only delivery-type orders have a delivery record
STAFF → DELIVERY	One-to-many	A staff member can handle many deliveries